Chapter 25

Correlated Strategies and Correlated Equilibrium

We commence our study of cooperative game theory with a discussion of games and solution concepts that enable us to capture cooperation among players. In this chapter, we first study *games with contracts* followed by *games with communication*. We introduce the notion of *correlated strategies* and show how these strategies capture cooperative actions of players. We then define and illustrate the notion of *correlated equilibrium* and bring out its connection to Nash equilibrium.

The motivation to look at games with cooperation is the fact that in many games, Nash equilibria yield non-optimal payoffs compared to certain non-equilibrium outcomes. Let us consider the following modified version of the prisoner's dilemma problem [1] whose payoff matrix is shown in Figure 25.1.

	2	
1	x_2	y_2
x_1	$2,2$	$0,6$
y_1	$6,0$	$1,1$

Fig. 25.1: Payoff matrix of a modified version of prisoner's dilemma game

Note in the above that the unique equilibrium (which happens to be a strictly dominant strategy equilibrium) is (y_1, y_2) that yields a payoff profile $(1,1)$. The non-equilibrium outcome (x_1, x_2) yields higher payoffs $(2,2)$. In situations like these, the players may like to transform the game to extend the set of equilibria to include better outcomes. There could be several ways of achieving this transformation:

- The players may communicate among themselves to coordinate their moves
- The players may formulate agreements in the form of contracts
- The players may decide to play the game repeatedly

25.1 Games with Contracts

In a game with contracts, a player who signs a contract is required to play according to a designated strategy which is called a correlated strategy. Contracts transform

games with less desirable equilibria to games with more desirable equilibria. We illustrate contracts and correlated strategies through the examples below.

Example 25.1 (Modified Prisoner's Dilemma with a Contract).
In the game shown in Figure 25.1, let us say the two players sign the following contract (call it contract 1).

(1) If both players sign this contract, then player 1 (player 2) chooses to play the strategy x_1 (x_2).
(2) If the contract is signed by only player 1, player 1 would choose y_1.
(3) If the contract is signed by only player 2, player 2 would choose y_2.

Call the action of signing the contract by player i ($i = 1, 2$) as a_i. We can now expand the strategy sets as $S_1 = \{x_1, y_1, a_1\}$ and $S_2 = \{x_2, y_2, a_2\}$. The transformed game has the payoff matrix shown in Figure 25.2.

	2		
1	x_2	y_2	a_2
x_1	$2, 2$	$0, 6$	$0, 6$
y_1	$6, 0$	$1, 1$	$1, 1$
a_1	$6, 0$	$1, 1$	$2, 2$

Fig. 25.2: Payoff matrix with contract 1

The transformed game now has a new equilibrium (a_1, a_2) which is a weakly dominant strategy equilibrium yielding payoffs $(2, 2)$. The profile (y_1, y_2) continues to be an equilibrium but is not a dominant strategy equilibrium. □

Example 25.2 (Modified Prisoner's Dilemma with Additional Contract).
Even better payoffs could be achieved if a second contract (call it contract 2) is introduced in addition to contract 1 above. This additional contract commits the players to a correlated strategy. This contract is as follows.

- If both players sign this new contract, then a coin will be tossed. In the event of a *heads*, they will implement (x_1, y_2) and in the event of a *tails*, they will implement (y_1, x_2).
- If player 1 alone signs this new contract, then player 1 chooses y_1.
- If player 2 alone signs this new contract, then player 2 chooses y_2.

If b_1 and b_2 represent the actions of players 1 and 2 corresponding to signing of this new contract, the extended payoff matrix would be as shown in Figure 25.3. This new game has the following equilibria:

- (y_1, y_2) with payoffs $(1, 1)$
- (a_1, a_2) with payoffs $(2, 2)$
- (b_1, b_2) with payoffs $(3, 3)$
- $((0, 0, \frac{2}{3}, \frac{1}{3}), (0, 0, \frac{2}{3}, \frac{1}{3}))$ where the mixed strategy $(0, 0, \frac{2}{3}, \frac{1}{3})$ for player 1 means a_1 with probability $\frac{2}{3}$ and b_1 with probability $\frac{1}{3}$. This equilibrium leads to a payoff of $\frac{5}{3}$ for both player 1 and player 2.

It turns out that none of the above equilibria are dominant strategy equilibria. □

	2			
1	x_2	y_2	a_2	b_2
x_1	$2,2$	$0,6$	$0,6$	$0,6$
y_1	$6,0$	$1,1$	$1,1$	$1,1$
a_1	$6,0$	$1,1$	$2,2$	$1,1$
b_1	$6,0$	$1,1$	$1,1$	$3,3$

Fig. 25.3: Modified prisoner's dilemma with contract 1 and contract 2

25.2 Correlated Strategies

The choice of pure strategies by players in a game may be correlated perhaps because the players are observing the same or related random events before making their choices. A correlated strategy captures the above feature. Moreover, a correlated strategy, as illustrated in the examples above, also captures cooperative play by the members of a coalition.

Definition 25.1 (Correlated Strategy). *Let $\Gamma = \langle N, (S_i), (u_i)\rangle$ be a strategic form game. A correlated strategy for a non-empty subset C (also called coalition) of the players is any probability distribution over the set of possible combinations of pure strategies that these players can choose.*

In other words, a correlated strategy, τ_C for a given coalition C belongs to $\Delta(S_C)$ where

$$S_C = \times_{i \in C} \ S_i$$

N is called the *grand coalition* and the symbol τ_N denotes a correlated strategy of the grand coalition.

Example 25.3 (Correlated Strategies). Let $N = \{1,2,3\}; S_1 = \{x_1, y_1\}; S_2 = \{x_2, y_2\}; S_3 = \{x_3, y_3, z_3\}$. If $C = \{2,3\}$, then

$$S_C = S_2 \times S_3 = \{(x_2, x_3), (x_2, y_3), (x_2, z_3), (y_2, x_3), (y_2, y_3), (y_2, z_3)\}$$

A correlated strategy for the coalition C is a probability distribution on S_C. For example, $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12})$ would correspond to (x_2, x_3) with probability $\frac{1}{4}$, (x_2, y_3) with probability $\frac{1}{4}$, and so on. □

Note. A correlated strategy $\tau_C \in \Delta(\times_{i \in C} S_i)$ can be implemented as follows. A reliable mediator or a referee picks randomly a profile of pure strategies in S_C according to distribution τ_C. The mediator asks each player to play the strategy chosen in this pure strategy profile.

Note. Let S, as usual, denote:

$$S = S_N = \times_{i \in N} \; S_i$$

Suppose $\alpha \in \Delta(S)$ is any correlated strategy for all players. Let $U_i(\alpha)$ denote the expected payoff to player i when α is implemented. It can be easily seen that

$$U_i(\alpha) = \sum_{s \in S} \alpha(s) u_i(s)$$

Let $U(\alpha) = (U_1(\alpha), \ldots, U_n(\alpha))$ denote the *expected payoff allocation* to players when they implement α.

Note. Given any allocation in the set $\{U(\alpha) : \alpha \in \Delta(S)\}$, there exists a contract such that if all the players signed this contract, then they would get this expected payoff allocation.

Note. The set of possible expected payoff allocations $\{U(\alpha) : \alpha \in \Delta(S)\}$ can be shown to be a *closed and convex* subset of $\mathbb{R}^n$. When there is no confusion, we will use $u(\alpha)$ instead of $U(\alpha)$.

Correlated Strategies and Mixed Strategies

It is important to understand the difference between a correlated strategy and a mixed strategy profile. A correlated strategy for the grand coalition N is a member of $\Delta(\times(S_i))$ whereas a mixed strategy profile is a member of $\times(\Delta(S_i))$. Given any mixed strategy profile σ of all players, we can always find a correlated strategy of all players α such that $u_i(\alpha) = u_i(\sigma)\ \forall i \in N$. However, given a correlated strategy α of all the players, it may not always be possible to find any mixed strategy profile σ such that $u_i(\sigma) = u_i(\alpha)\ \forall i \in N$.

Contract Signing Game

A vector of correlated strategies of all possible coalitions is called a *contract*. More formally, a contract is defined as follows.

Definition 25.2 (Contract). *Consider the vector $\tau = (\tau_C)_{C \subseteq N}$. Note that*

$$\tau \in \times_{C \subseteq N} \; \left(\Delta(\times_{i \in C} \; S_i)\right).$$

The vector τ of correlated strategies of all coalitions is a contract.

Note that τ_C for $C \subseteq N$ gives the correlated strategy that would be implemented by players in C if C were the set of players to sign the contract. Clearly, a contract defines (in fact induces) an extended game and this extended game is called the *contract signing game*.

Example 25.4 (Contract in terms of Correlated Strategies). We have discussed two contracts in the context of the modified prisoner's dilemma. Note that contract 1 is described by $(\tau_1, \tau_2, \tau_{\{1,2\}})$ where $\tau_1 = (x_1 : 0, y_1 : 1)$; $\tau_2 = (x_2 : 0, y_2 : 1)$; $\tau_{\{1,2\}} = ((x_1, x_2) : 1;\ (x_1, y_2) : 0;\ (y_1, x_2) : 0;\ (y_1, y_2) : 0)$. The payoff matrix in Figure 25.2 defines the contract signing game induced by contract 1. Contract 2 is given by $(\tau_1, \tau_2, \tau_{\{1,2\}})$ where $\tau_1 = (x_1 : 0, y_1 : 1)$; $\tau_2 = (x_2 : 0, y_2 : 1)$; $\tau_{\{1,2\}} = ((x_1, x_2) : 0;\ (x_1, y_2) : \frac{1}{2};\ (y_1, x_2) : \frac{1}{2};\ (y_1, y_2) : 0)$. The payoff matrix in Figure 25.3 defines the contract signing game induced by contract 1 and contract 2. □

Concept of Individual Rationality

Clearly, players may not be willing to sign any contract that is proposed. For example, in the modified prisoner's dilemma game of Figure 25.1, player 1 will not agree to sign a contract that would commit the players to implement (x_1, y_2) since it gives him a payoff 0. Player 1 can always guarantee himself a payoff of 1 by not signing anything and simply choosing y_1. We now investigate which contracts a player would be really interested in signing. In Chapters 6 and 7, we have already seen the notion of a *security level* or *maxmin value* for a player, which is the minimum guaranteed payoff the player can assure himself when the rest of the players are free to play any strategies. The following provides a natural way of defining such a security level:

$$v_i = \max_{\tau_i \in \Delta(S_i)} \min_{s_{N-i} \in S_{N-i}} u_i(\tau_i, s_{N-i})$$

where

$$S_{N-i} = S_{N \setminus \{i\}}; \quad s_{N-i} \in S_{N-i}.$$

The expected utility $u_i(\tau_i, s_{N-i})$ is given by:

$$u_i(\tau_i, s_{N-i}) = \sum_{s_i \in S_i} \tau_i(s_i) u_i(s_i, s_{N-i}).$$

For example, in the modified prisoner's dilemma problem (Figure 25.1), $v_1 = v_2 = 1$.

Individually Rational Correlated Strategy

It is reasonable for player i to sign a contract to play a correlated strategy α only if $u_i(\alpha) \geq v_i$. This is called the *individual rationality* or *participation* constraint for player i. This leads to the following definition.

Definition 25.3 (Individually Rational Correlated Strategy). *A correlated strategy*

$$\alpha \in \Delta(S_1 \times \cdots \times S_n)$$

for all the players in N is said to be individually rational *if*

$$u_i(\alpha) \geq v_i \quad \forall i \in N.$$

It can be shown, using a two player zero-sum game interpretation of the above situation, that the security level v_i also satisfies

$$v_i = \min_{\tau_{N-i} \in \Delta(S_{N-i})} \max_{s_i \in S_i} \; u_i(s_i, \tau_{N-i})$$

where

$$\tau_{N-i} = \tau_{N \setminus \{i\}} \in \Delta(S_{N-i}).$$

The expected utility $u_i(s_i, \tau_{N-i})$ is given by

$$u_i(s_i, \tau_{N-i}) = \sum_{s_{N-i} \in S_{N-i}} \tau_{N-i}(s_{N-i}) u_i(s_i, s_{N-i}).$$

v_i is the best expected payoff that player i is guaranteed to get against any correlated strategy that the other players could use against player i. A *minmax correlated strategy* against player i is any correlated strategy $\tau_{N-i} \in \Delta(S_{N-i})$ of the rest of the players which forces the payoff of player i to be equal to his minmax value v_i.

Equilibria of the Contract Signing Game

Suppose the players make their decisions about which contract to sign independently. Then, the following proposition holds.

Proposition 25.1. *Given any individually rational correlated strategy α, there exists a contract τ with $\tau_N = \alpha$ such that all players signing this contract is a Nash equilibrium of the contract signing game.*

Proof: Let $\alpha \in \Delta(S)$ be an individually rational correlated strategy. That is

$$u_i(\alpha) \geq v_i \;\; \forall i \in N$$

Consider the contract $\tau = (\tau_C)_{C \subseteq N}$ such that $\tau_N = \alpha$ and τ_{N-i} is a minmax strategy against player i. τ_C for all other coalitions (that is, coalitions other than $N, N-1, \ldots, N-n$) could be chosen arbitrarily. Let $(a_1, \ldots, a_n)$ be the profile of contract signing strategies for this contract. Note that this profile corresponds to the situation when all the players sign the contract. Now,

$$u_i(a_1, \ldots, a_n) = u_i(\alpha) \quad (\text{since } \tau_N = \alpha).$$

Therefore, we get for all $i \in N$,

$$u_i(a_1, \ldots, a_n) \geq v_i$$

For any $s_i \in S_i$ such that $s_i \neq a_i$, note that the profile (s_i, a_{-i}) corresponds to the situation when player i plays s_i and the rest of the players sign the contract. We have,

$$u_i(s_i, a_{-i}) = u_i(s_i, \tau_{N-i}) \;\leq v_i$$

since τ_{N-i} is a minmax strategy against player i. Therefore,

$$u_i(a_i, a_{-i}) \geq u_i(s_i, a_{-i}) \;\; \forall s_i \in S_i \;\; \forall i \in N$$

Thus $(a_1, a_2, \ldots, a_n)$ is a Nash equilibrium of the contract signing game. ■

Thus any individually rational correlated strategy $\alpha \in \Delta(S)$ will lead to a contract such that all players signing the contract is a Nash equilibrium of the contract signing game. It also turns out that in any Nash equilibrium of a contract signing game, the expected payoffs for players would be at least their respective security levels. This is stated in the next proposition.

Proposition 25.2. *Consider any Nash equilibrium (s_i^*, s_{-i}^*) of a contract signing game induced by a correlated strategy α. Then, $u_i(s_i^*, s_{-i}^*) \geq v_i \; \forall i \in N$.*

Proof: Consider any Nash equilibrium (s_i^*, s_{-i}^*) of a contract signing game and suppose $u_i(s_i^*, s_{-i}^*) < v_i$ for some player i. Now player i can decide not to sign the contract and instead play the strategy that guarantees him the minmax value v_i. This immediately provides the contradiction. ■

Following the two propositions above, it is clear that

$$\{u(\alpha) : \alpha \in \Delta(S) \text{ and } \quad u_i(\alpha) \geq v_i \; \forall i \in N\}$$

is exactly the set of payoff allocations that can be achieved in Nash equilibria of the contract signing game when every player has the option to sign nothing and choose an action in S_i. This set is also the set of expected payoff allocations corresponding to individually rational correlated strategies. This set can be shown to be *closed and convex*.

Example 25.5 (Payoff Allocations in the Modified Prisoner's Dilemma). Consider the modified prisoner's dilemma game with payoff matrix as in Figure 25.1. The expected payoff in a correlated strategy is a convex combination of the payoffs in different strategy profiles and hence the set of all payoff allocations for the above game is the convex set with extreme points $(0, 6)$, $(6, 0)$, and $(1, 1)$. See Figure 25.4. Note that this set is also closed and that the point $(2, 2)$ is in the interior of this convex set. The set of possible expected payoff allocations satisfying individual rationality is the triangle with corners at $(1, 1)$, $(5, 1)$, and $(1, 5)$ as shown. This is because $v_1 = 1$ and $v_2 = 1$ for this example. Note that this set is also closed and convex. □

25.3 Games with Communication

We have so far seen how contracts can transform a game with less desirable equilibria to a game with more desirable equilibria. However, in many situations, players may not be able to commit themselves to binding contracts. The reasons for this could be many [1]:

- Player's strategies may not be observable to the enforcers of contracts.
- There may not be adequate sanctions to guarantee compliance with contracts.

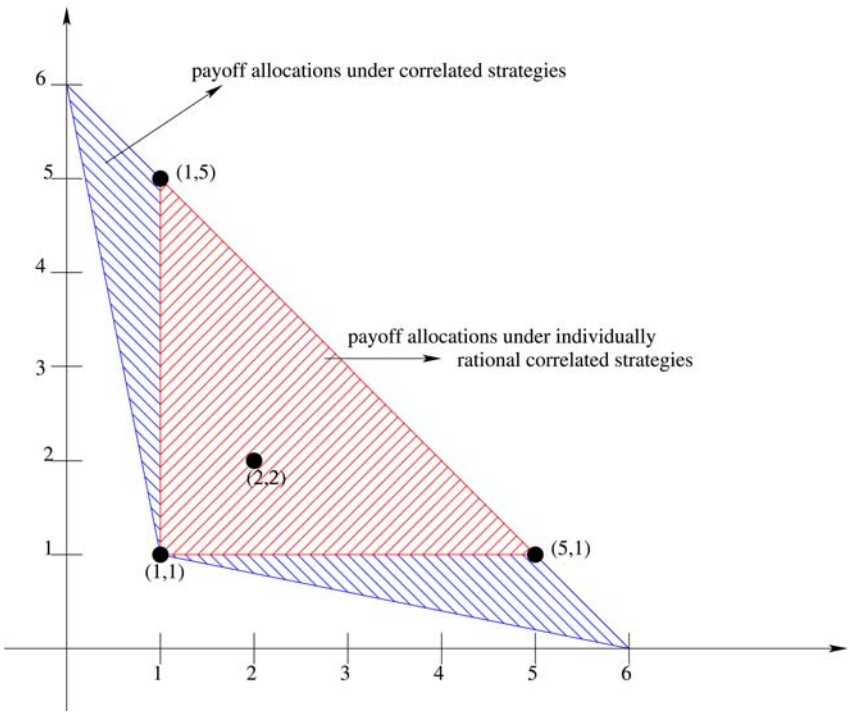

Fig. 25.4: Payoff allocation vectors under correlated strategies for the modified prisoner's dilemma problem

- Player's strategies might actually involve inalienable rights (that is, players may not be able to transfer their rights).

In these above situations also, it may still be possible for the players to communicate and coordinate among themselves, and achieve a self-enforcing equilibrium with desirable payoff structures. Such games correspond to *games with communication*. A game with communication is one, in which, in addition to the strategy options explicitly specified, the players have a range of implicit options to communicate with each other. A game with communication need not have any contracts. Such a game may still achieve interesting results in spite of contracts being absent.

Example 25.6 (A Game with Communication). Consider a two player game having the payoff matrix shown in Figure 25.5. Clearly, this game has three Nash equilibria.

- (x_1, x_2) with payoff allocation $(5, 1)$
- (y_1, y_2) with allocation $(1, 5)$
- Mixed strategy Nash equilibrium $((\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}))$ which yields the outcome $(2.5, 2.5)$.

Note that (y_1, x_2) is not a Nash equilibrium though it is clearly a desirable outcome. It can be realized through a binding contract. However, if contracts cannot be used, we are not sure whether we can achieve the payoff profile $94, 4)$ or even anything better than

	2	
1	x_2	y_2
x_1	$5,1$	$0,0$
y_1	$4,4$	$1,5$

Fig. 25.5: A two player game to illustrate games with communication

$(2.5, 2.5)$. It turns out that correlated strategies exist that achieve a better allocation than $(2.5, 2.5)$ as shown by the following example. □

Example 25.7 (Correlated Strategy 1). Let us say players choose to toss a coin and select the outcome (x_1, x_2) with probability $\frac{1}{2}$ and the outcome (y_1, y_2) with probability $\frac{1}{2}$. This refers to the following correlated strategy

$$\alpha = \left((x_1, x_2); \frac{1}{2}; (x_1, y_2) : 0; (y_1, x_2) : 0; (y_1, y_2) : \frac{1}{2}\right)$$

Note that $u_1(\alpha) = u_2(\alpha) = 3$. To implement this correlated strategy, players can toss a coin and choose the outcome (x_1, x_2) in the event of a *heads* and choose the outcome (y_1, y_2) in the event of a *tails*. In order to implement this, communication and coordination are required. The outcome suggested by this correlated strategy will be *self-enforcing* since neither player will gain by unilaterally deviating from this.

The above correlated strategy can also be implemented with the help of a trusted mediator who helps the players to communicate and share information: the mediator recommends, randomly, with probability 0.5 each the profiles (x_1, x_2) and (y_1, y_2). Assume that each player comes to know only the strategy recommended to him by the mediator.

- Player 1, if recommended x_1 by the mediator thinks that player 2 is recommended x_2. Believing that player 2 obeys the recommendation x_2, player 1 finds it a best response to choose x_1 and thus accepts the mediator's recommendation.
- Player 1, if recommended y_1 by the mediator thinks that player 2 is recommended y_2. Believing that player 2 obeys the recommendation y_2, player 1 finds it a best response to choose y_1 and again accepts the mediator's recommendation.
- Player 2, if recommended x_2 by the mediator thinks that player 1 is recommended x_1. Believing that player 1 obeys the recommendation x_1, player 2 finds it a best response to choose x_2 and thus accepts the mediator's recommendation.
- Player 2, if recommended y_2 by the mediator thinks that player 1 is recommended y_1. Believing that player 1 obeys the recommendation y_1, player 2 finds it a best response to choose y_2 and again accepts the mediator's recommendation.

Thus mediation can also be used to implement the above correlated strategy. □

Example 25.8 (Correlated Strategy 2). We now explore a different correlated strategy that can be realized with the help of a mediator. Consider the correlated strategy

$$\alpha = \left((x_1, x_2) : \frac{1}{3};\ (x_1, y_2) : 0;\ (y_1, x_2) : \frac{1}{3};\ (y_1, y_2) : \frac{1}{3}\right)$$

Note that $u_1(\alpha) = u_2(\alpha) = \frac{10}{3}$. To implement this correlated strategy, the mediator recom-

mends, randomly, with probability $\frac{1}{3}$, each of the profiles $(x_1, x_2), (y_1, y_2), (y_1, x_2)$. Again assume that each player comes to know only the strategy recommended to him by the mediator.

- Suppose the mediator recommends x_1 to player 1. Then player 1 knows that player 2 is recommended x_2. When player 2 plays x_2 (as recommended by the mediator), it is a best response for player 1 to play x_1, so he would be happy to play x_1 and thus accept the recommendation of the mediator.
- Suppose the mediator recommends y_1 to player 1. Then player 1 knows that the mediator would recommend the mixed strategy $x_2 : 0.5; y_2 : 0.5$ to player 2. When player 2 plays the above mixed strategy, then player 1 gets a payoff of 2.5 if he plays x_1 and gets a payoff of 2.5 even if he plays y_1. Thus player 1 can be indifferent between x_1 and y_1 and will not mind accepting the recommendation of the mediator to play y_1.

The above shows that player 1 would be happy to follow the recommendation of the mediator if player 1 expected player 2 also to follow the recommendation of the mediator.

Similarly, it can be shown that player 2 would be happy to do as recommended by the mediator under the belief that player 1 would do as recommended by the mediator.

The above shows that the two players can reach a self-enforcing understanding to obey the mediator if the mediator recommends the correlated strategy

$$\left((x_1, x_2) : \frac{1}{3};\ (x_1, y_2) : 0;\ (y_1, x_2) : \frac{1}{3};\ (y_1, y_2) : \frac{1}{3}\right)$$

In other words, even though the mediator's recommendation is not binding on the two players, the two players find it in their best interest to follow it. Thus there is a Nash equilibrium of the transformed game with mediated communication without contracts. This is the idea behind the notion of *correlated equilibrium* which is discussed next. □

25.4 Correlated Equilibrium

Consider the following setup of a game with communication. Let $\Gamma = \langle N, (S_i), (u_i)\rangle$ be any finite strategic form game. Assume that there is a mediator who chooses a pure strategy profile $(s_1, \ldots, s_n)$ according to a probability distribution $\alpha \in \Delta(S)$ which is common knowledge among the players. The observer recommends to player i $(i = 1, \ldots, n)$ the pure strategy s_i but does not reveal s_{-i} to player i. Based on the recommendation, the player either obeys it or chooses any other strategy from his strategy set S_i. Let $\delta_i : S_i \to S_i$ describe player i's choice of a strategy based on the mediator's recommendation. That is, $\delta_i(s_i)$ gives the strategy that the player i chooses to play when the mediator recommends s_i to him. $\delta_i(s_i) = s_i$ means that the player i obeys the mediator when the mediator recommends s_i.

Definition 25.4 (Correlated Equilibrium). *Given a finite strategic form game $\Gamma = \langle N, (S_i), (u_i)\rangle$, a correlated strategy $\alpha \in \Delta(S)$ recommended by a mediator is called a correlated equilibrium if*

$$u_i(\alpha) \geq \sum_{(s_i, s_{-i}) \in S} \alpha(s_i, s_{-i})\ \ u_i(\delta_i(s_i), s_{-i})\ \ \forall \delta_i : S_i \to S_i\ \ \forall i \in N. \tag{25.1}$$

Note that when a correlated equilibrium is recommended, every player finds it a best response to obey the recommendation of the mediator. Thus a correlated equilibrium is any correlated strategy for the players which could be self-enforcingly implemented with the help of a mediator who can make non-binding recommendations to each player.

Computing Correlated Equilibria

It can be shown that the inequalities (25.1) are equivalent to the following set of inequalities.

$$\sum_{s_{-i} \in S_{-i}} \alpha(s)[u_i(s_i, s_{-i}) - u_i(s_i', s_{-i})] \geq 0 \quad \forall s_i \in S_i;\ \forall s_i' \in S_i;\ \forall i \in N. \tag{25.2}$$

The equivalence can be shown by fixing s_i and unfolding the original inequalities (left as an exercise). Equation (25.2) asserts that no player i could expect to increase his expected payoff by using some disobedient action s_i' when the mediator recommends s_i. The constraints (25.2) are called *strategic incentive constraints*. They are the constraints to be satisfied by a mediator's correlated strategy for ensuring that all players could rationally obey the recommendations. It can also be noted that

$$\alpha(s) \geq 0,\ \forall s \in S$$
$$\sum_{s \in S} \alpha(s) = 1$$

Note. It can be shown that the set of all payoff allocations under correlated equilibria in a finite game is a *compact and convex* set. Recall that the set of all payoff allocations under correlated strategies as well as the set of payoff allocations under individually rational correlated strategies are also compact and convex.

Let us look at the following linear program:

$$\max \sum_{i \in N} u_i(\alpha)$$

subject to

$$\sum_{s_{-i} \in S_{-i}} \alpha(s) \left[u_i(s_i, s_{-i}) - u_i(s_i', s_{-i})\right] \geq 0 \quad \forall i \in N\ \forall s_i \in S_i\ \forall s_i' \in S_i$$
$$\alpha(s) \geq 0\ \ \forall s \in S$$
$$\sum_{s \in S} \alpha(s) = 1$$

Any feasible solution of this linear program will give a correlated equilibrium. An optimal solution of this linear program will give a correlated equilibrium that maximizes the social welfare.

Example 25.9 (Computation of correlated equilibria). Consider the game shown in Figure 25.5. The linear program here is:

$$\text{maximize } 6\,\alpha(x_1,x_2) + 0\,\alpha(x_1,y_2) + 8\,\alpha(y_1,x_2) + 6\,\alpha(y_1,y_2) \quad \text{subject to}$$

$$\begin{aligned}
(5-4)\,\alpha\,(x_1,x_2) + (0-1)\,\alpha\,(x_1,y_2) &\geq 0\\
(4-5)\,\alpha\,(y_1,x_2) + (1-0)\,\alpha\,(y_1,y_2) &\geq 0\\
(1-0)\,\alpha\,(x_1,x_2) + (4-5)\,\alpha\,(y_1,x_2) &\geq 0\\
(0-1)\,\alpha\,(x_1,y_2) + (5-4)\,\alpha\,(y_1,y_2) &\geq 0\\
\alpha\,(x_1,x_2) + \alpha\,(x_1,y_2) + \alpha\,(y_1,x_2) + \alpha\,(y_1,y_2) &= 1
\end{aligned}$$

This yields the solution

$$\alpha\,(x_1,x_2) = \frac{1}{3};\ \alpha\,(y_1,x_2) = \frac{1}{3};\ \alpha\,(y_1,y_2) = \frac{1}{3};\ \ \alpha(x_1,y_2) = 0$$

This shows that the above correlated strategy yields the maximum total payoff to the two players. The above also shows that the sum of expected payoffs cannot exceed $6\frac{2}{3}$ under non-binding mediated communication scenario.

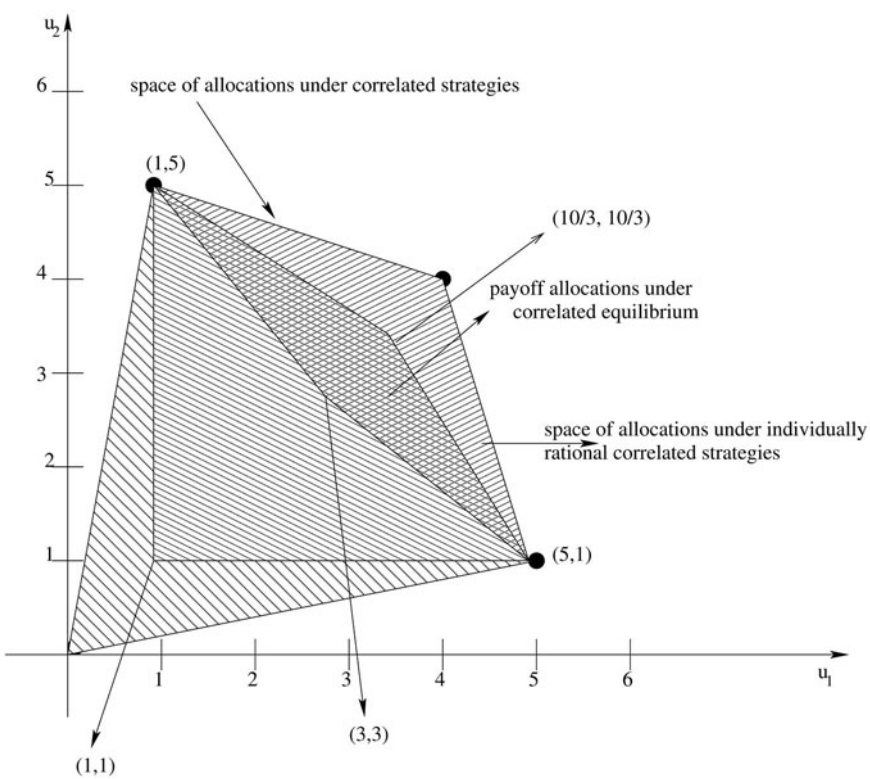

Fig. 25.6: Payoff allocation vectors under correlated strategies and correlated equilibria

Figure 25.6 shows the sets of all payoff allocations for this example under correlated strategies (convex hull with corner points $(0,0),(1,5),(5,1),(4,4)$), under individually rational correlated strategies (convex hull with corner points $(1,1),(1,5),(5,1),(4,4)$) , and under correlated equilibria (convex hull with corner points $(1,5),(5,1),(3,3),(\frac{10}{3},\frac{10}{3})$). $\square$

We can now ask the following question: Can we select a small number of (perhaps a single) desirable or best outcomes among the above sets. For two player games, this issue was settled by the Nash bargaining theorem which we discuss in the next chapter. For multi-player games (including two player games), a variety of solution concepts have been suggested: The core, Shapley value, bargaining sets, nucleolus, etc. We will be studying these in the other following chapters.

25.5 Summary and References

The key learnings from this chapter include the following.

- Games with contracts and games with communication provide different ways of modeling cooperation among players. Contracts and communication transform games with less desirable equilibria to games with more desirable equilibria. Correlated strategies provide the framework for defining games with contracts and games with communication.
- Given a coalition of players, a correlated strategy of the players is a probability distribution on the space of strategy profiles of these players. A correlated strategy captures cooperation among the players in that coalition. A correlated strategy also enables to express the fact that the choice of pure strategies by players may be correlated since they might be observing the same or similar random events in deciding which pure strategies to play.
- A correlated strategy of the grand coalition is said to be individually rational if the expected payoff for each player under this correlated strategy is at least the security value or minmax value of the player. An individually rational correlated strategy induces a Nash equilibrium in the contract signing game.
- A correlated equilibrium is a correlated strategy of the grand coalition which leads to all players self-enforcingly obeying the recommendations of a mediator who implements the correlated strategy.
- The set of all correlated equilibria could be computed as feasible solutions of a linear program. Using this linear program, we can also compute correlated equilibria that maximize social welfare.
- The following sets of payoff allocations are convex and closed. (1) Set of all payoff allocations under correlated strategies (2) Set of all payoff allocations under individually rational correlated strategies which is the same as the set of allocations under Nash equilibria of contract signing games (3) Set of all payoff allocations under correlated equilibria.

The material discussed in this chapter draws upon mainly from the book by Myerson [1]. The concept of correlated equilibrium was first introduced by Robert Aumann [2]. There are many recent papers on efficient algorithms for computing correlated equilibria. The reader is referred to the book by Chalkiadakis, Elkind, and Wooldridge [3]. For an excellent detailed treatment of correlated strategies and correlated equilibria, we refer the reader to the books by Peleg and Sudholter [4] and Maschler, Solan, and Zamir [5].

References

[1] Roger B. Myerson. *Game Theory: Analysis of Conflict.* Harvard University Press, Cambridge, Massachusetts, USA, 1997.

[2] Robert J. Aumann. "Subjectivity and correlation in randomized strategies". In: *Journal of Mathematical Economics* **1** (1974), pp. 67–95.

[3] Georgios Chalkiadakis, Edith Elkind, and Michael Wooldridge. *Computational Aspects of Cooperative Game Theory.* Morgan & Claypool, 2011.

[4] B. Peleg and P. Sudholter. *Introduction to the Theory of Cooperative Games.* Kluwer Academic, Boston, USA, 2003.

[5] Michael Maschler, Eilon Solan, and Shmuel Zamir. *Game Theory.* Cambridge University Press, 2013.

25.6 Exercises

(1) Given a strategic form game and a mixed strategy profile σ, find a correlated strategy α such that $u_i(\alpha) = u_i(\sigma) \ \ \forall i \in N$.

(2) Suppose $(\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed strategy Nash equilibrium of a strategic form game $\Gamma = \langle N, (S_i), (u_i) \rangle$. Define a correlated strategy induced by the above equilibrium:

$$\alpha(s_1, \ldots, s_n) = \sigma_1^*(s_1) \ldots \sigma_n^*(s_n); \ \ \forall s_i \in S_i \ \ \forall i = 1, \ldots, n.$$

Show that such a correlated strategy is a correlated equilibrium.

(3) If α is a correlated equilibrium of a strategic form game, show that $u_i(\alpha)$ is equal to the maxmin value of i for each $i = 1, \ldots, n$.

(4) If α is a correlated equilibrium of a matrix game, show that $u_1(\alpha)$ is equal to the value in mixed strategies of the game.

(5) Give a simple example of a strategic form game such that the space of all payoff profiles achievable under mixed strategy profiles is the same as the space of all payoff profiles achievable under correlated strategies.

(6) Give a simple example of a strategic form game such that the space of all payoff profiles achievable under mixed strategy profiles is a strict subset of the space of all payoff profiles achievable under correlated strategies.

(7) Given a strategic form game, define the minmax value under correlated strategies

as:

$$v_i = \min_{\tau_{N-i} \in \Delta(S_{N-i})} \max_{s_i \in S_i} u_i(s_i, \tau_{N-i})$$

Show, using a two player zero-sum game interpretation of the above game, that the value v_i also satisfies

$$v_i = \max_{\tau_i \in \Delta(S_i)} \min_{s_{N-i} \in S_{N-i}} u_i(\tau_i, s_{N-i})$$

where

$$u_i(\tau_i, s_{N-i}) = \sum_{s_i \in S_i} \tau_i(s_i) u_i(s_i, s_{N-i})$$

(8) Given a finite strategic form game, show that the following sets are closed and convex.

- The space of all payoff allocations achievable under correlated strategies
- The space of all payoff allocations achievable under individually rational correlated strategies
- The space of all payoff allocations achievable under correlated equilibria

(9) Show the equivalence of the set of inequalities presented in equation (25.2) and the set of inequalities presented in equation (25.1).

(10) Compute all correlated equilibria of

	2	
1	x_2	y_2
x_1	$2,2$	$0,6$
y_1	$6,0$	$1,1$

	2	
1	x_2	y_2
x_1	$2,2$	$0,0$
y_1	$0,0$	$1,1$

(11) Consider the following two player game:

	A	B
A	$5,5$	$1,1$
B	$1,1$	$3,3$

For this game,

- compute the set of all payoff utility pairs possible (a) under correlated strategies (b) under individually rational correlated strategies.
- compute all correlated equilibria that maximize the sum of utilities of the two players.

(12) Consider the following two player game:

	A	B
A	$4,1$	$0,0$
B	$3,3$	$1,4$

For the above game, compute:

- the space of all payoff allocations under mixed strategy profiles
- the space of all payoff allocations under mixed strategy Nash equilibria
- the space of all payoff allocations under correlated strategies
- the space of all payoff allocations under individually rational correlated strategies
- the space of all payoff allocations under correlated equilibria

(13) Suppose $\sigma^* = (\sigma_1^*, \ldots, \sigma_n^*)$ is a mixed strategy Nash equilibrium of a strategic form game. Consider the correlated strategy $\alpha_{\sigma^*} \in \Delta(S)$ defined by

$$\alpha_{\sigma^*}(s_1, \ldots, s_n) = \sigma_1^*(s_1) \times \cdots \times \sigma_n^*(s_n).$$

Show that the above correlated strategy is a correlated equilibrium.

(14) **Programming Assignment**. Computing correlated equilibria of a finite strategic form game is an important computational problem. We have shown that these can be computed as feasible solutions of a linear program. Design and develop a program to compute correlated equilibria and welfare maximizing correlated equilibria. Also, compute the spaces of allocations under correlated strategies, under individually rational correlated strategies, and under correlated equilibria.